The 120 V side powers the lower-power auxiliary components and is kept separate from the 480 V motor system. The **120 VAC outlet first passes through overcurrent protection and a GFCI**, providing both short-circuit protection and additional protection against ground faults. The 120 V supply is then distributed directly to the **120 V indicator light and fan**, since neither requires voltage conversion. A third branch feeds a **120 VAC → 24 VDC power supply**, which converts the available AC power into the regulated 24 VDC required by the IMU and gas sensor.

One change I would make is to ensure the **GFCI and breaker aren't presented as doing the same job**: the breaker protects the branch/wiring against excessive current, while the GFCI detects current leaking through an unintended path to ground. I would also include **individual branch protection where appropriate** after distribution, particularly for the 24 V power supply. Otherwise, the overall architecture is solid: **120 VAC → protection → distribution → 120 V loads + 24 VDC conversion → 24 V sensors.**

